

Gas Leak Detector (GLD) Instruction Manual

IoT-based multi-gas leak detection node — GLD Tahap 2 program, PT Pertamina Patra Niaga

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Object of this manual	Node Sensor (GLD) only — Cluster Head and Gateway are separate devices, referenced only for system context
Related certification item	IECEx/ATEX checklist §2.6.g — Usage and installation instructions

Scope and status of this document.

This is a working draft, modeled structurally on a published instruction manual for a comparable certified diffusion-type gas detector, and populated with GLD's own confirmed design data wherever available. It is **not** an approved or finalized manual, and GLD is **not yet ATEX/IECEx certified** — see Section 15, Certification Status. Fields not yet established are explicitly marked **Pending confirmation** rather than estimated. This document must be reviewed and approved by LGU engineering and, where applicable, by Pertamina RU IV before being treated as an operational instruction.

Revision	Date	Description
0.1	22 Sep 2026	Initial working draft, structured from the reference manual of a comparable certified gas detector (New Cosmos KD-12/KD-12R) and populated with GLD's confirmed design data.

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1. Introduction

The GLD (Gas Leak Detector) is an IoT-based, multi-gas leak detection sensor node developed by LAPI Ganesha Utama (LGU) together with the IoT Lab and Physics Lab of Institut Teknologi Bandung (ITB), under the GLD Tahap 2 program for PT Pertamina Patra Niaga. GLD is intended to detect gas leakage at an early stage in refinery facilities and to report gas concentration status wirelessly to a central monitoring dashboard, helping to prevent incidents such as fires and explosions.

Unlike a conventional loop-powered gas detector with a local 4–20 mA analog output and an on-unit keypad/display, GLD is a wireless sensor node: it classifies gas conditions on-device and reports status over a LoRa mesh network (via a Cluster Head and Gateway) to a central server and dashboard. GLD has no physical keypad or digital concentration display; status is conveyed through a local visual/audible alarm module and remotely through the dashboard.

GLD currently classifies ambient air into three conditions on-device: **Clean Air, LPG, and H₂** (hydrogen). It does not yet cover CO₂, Benzene, CO, or H₂S; expanding gas coverage is an active area of the program (see Section 13, Specifications).

GLD is not yet certified for use in ATEX/IECEx classified hazardous areas. Certification is in progress — see Section 14, Certification Status. Do not install GLD in a classified hazardous area until certification is complete and the equipment has been confirmed suitable for the applicable zone, gas group, and temperature class.

2. Safety Precautions

WARNING

- Do not install GLD in a classified (ATEX/IECEX) hazardous area until certification of the equipment for the applicable zone, gas group, and temperature class has been completed and confirmed. GLD's certification is currently in progress (Section 14).
- Any cable entry device and blanking element used with GLD in a hazardous area must be of a type of protection and certification suitable for the intended zone and correctly installed. GLD's own cable entry specification is currently **Pending confirmation** (Section 13); do not assume a specific cable gland is suitable without written confirmation.
- Do not disassemble the product or modify its enclosure, wiring, or firmware. Doing so may compromise any explosion-protection measures once certification is finalized, and will void any future certification coverage.
- Be sure to ground the unit in accordance with the site's electrical safety requirements to prevent electric shock.

CAUTION

- All work involving GLD — including installation, wiring, and mounting — must be carried out by suitably trained personnel in accordance with the applicable code of practice and site HSE procedures.
- Do not install GLD near sources of silicone sealant vapor or other substances known to poison metal-oxide gas sensors; this may permanently degrade sensor performance.
- Handle the sensor unit and enclosure with care. The gas-sensing elements are directly exposed to ambient air (not enclosed) and can be damaged by physical impact, dust ingress, or contact with liquids.
- Mounting to existing plant structures must not involve drilling or welding unless a different method has been separately approved in writing by the site (RU IV). See Section 5, Installation.
- Fasteners used for mounting must use a material and grade suitable for the site environment (anti-corrosion required); the specific fastener grade for GLD's bracket is currently **Pending confirmation** (Section 13).

NOTE

GLD's gas-sensing elements (metal-oxide type) are intentionally exposed to ambient air through the enclosure mesh, in the same manner as comparable diffusion-type gas detectors, in order to sense gas concentration. The ignition-protection status of this sensing aperture (e.g. whether a certified flame-arrestor element is required) is under evaluation as part of the ongoing ATEX/IECEX certification process and is not yet finalized.

3. Package Contents

The following items are expected to accompany each GLD unit at delivery. This list reflects the current project design and will be finalized alongside the certification and vendor handoff process.

ITEM	STATUS
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GLD sensor node (main unit)	Confirmed
Mounting bracket set (L-bracket + U-bolt, 2"/DN50)	Confirmed — see Section 5
External LoRa/Wi-Fi antenna (SMA connector)	Confirmed
Power adapter (220 VAC/50 Hz → 24 VDC) or site-supplied 24 VDC feed	Partially confirmed — supply arrangement to be agreed with site (Section 6)
Cable gland(s)	Pending confirmation — specification not yet determined (Section 13)
Fasteners for bracket mounting	Pending confirmation — grade not yet determined (Section 13)
This Instruction Manual	Confirmed

4. External Dimensions and Nomenclature

Overall dimensions: **200 × 90 × 290 mm** (L×W×H). Enclosure neck diameter **Ø75 mm**, probe/casing height **191.51 mm**. Weight: **≈2 kg** (project-confirmed value; not yet verified against a calibrated weighing record). Enclosure material: aluminum alloy and stainless steel (specific alloy/grade **Pending confirmation**).



Front view — assembled unit: motherboard housing (blue), sensor mesh cover (silver, top), external antenna (left), local visual/audible alarm module (right).



Rear view — mounting flange (4-hole pattern), matching the L-bracket/U-bolt mounting plate described in Section 5.

COMPONENT	DESCRIPTION
Sensor cover / mesh	Perforated cover over the 8-channel gas sensor array; sensing elements are directly exposed to ambient air (not enclosed).

Local alarm module	Cylindrical housing with red lens, mounted to one side of the enclosure; provides local visual (flashing) and audible alarm indication.
External antenna	External, omnidirectional, SMA male connector, 3 dBi gain, used for the LoRa radio link. On some units the 2.4 GHz Wi-Fi antenna remains inside the enclosure and must be relocated externally for optimal configuration-mode range.
Cable entry	Location for power and (where used) RS-485 wiring. Thread size and certified cable gland requirement: Pending confirmation .
Mounting flange	4-hole pattern on the rear of the enclosure, matched to the L-bracket mounting plate (Section 5).
Status indication	No on-unit digital display or keypad. Status is conveyed via the local alarm module and remotely via the dashboard; see Section 10.

5. Installation

5.1 Mounting Method

GLD is mounted to existing plant structures using an L-bracket and U-bolt clamp, without drilling or welding into the structure. This design follows the reference approach already used for corrosion-monitoring instrumentation at the refinery (Emerson-style bracket).

PARAMETER	VALUE	STATUS
Clamp type	U-bolt, 2" / DN50	Final
Target pipe / structure outer diameter	60.3 mm	Final
U-bolt thread	M10	Final
U-bolt inside width	62–65 mm	Final
U-bolt inside height	95–100 mm	Final
Thread length per leg	40 mm recommended (35 mm minimum viable)	Final
Mounting plate	250 × 250 mm	Final
Spacer (if bracket/structure is thicker)	45–60 mm	Final
Mounting orientation	Confirmed suitable for horizontal handrail-type structures. Compatibility with a vertical pole has not yet been verified in writing .	Pending confirmation

Fastener size and grade	Not yet determined by the design team. Generic industry reference standards (not an official GLD spec): BS 3974 Part 2, DIN 3570/3567, ASME B18.31.2/ASTM A193 Gr.B7.	Pending confirmation
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CAUTION

- Do not drill or weld the existing structure. Use only the L-bracket/U-bolt method described above unless an alternative method has been separately approved in writing by RU IV.
- PVC must not be used in any housing or bracket component. All fasteners and brackets must be of an anti-corrosion material suitable for the site environment.
- Confirm the mounting orientation (horizontal vs. vertical structure) with the installing party before fabrication, as vertical-pole compatibility is not yet written-verified.

5.2 Mounting Location and Height

GLD is designed to detect multiple gases with different vapor densities (for example, LPG is heavier than air, while H₂ is lighter than air). Unlike a single-gas detector, a single mounting height cannot simultaneously be optimal for every gas GLD may be asked to detect. The mounting height and location for each installation point should therefore be selected by site engineering based on the specific target gas(es) and hazard assessment for that location, rather than a single fixed rule.

General siting guidance (adapted from established gas-detector practice, to be applied per site hazard assessment):

RELATIVE GAS DENSITY	GENERAL GUIDANCE
Heavier than air (e.g., LPG)	Mount low, close to grade or process equipment where the gas is expected to accumulate.
Similar to air (e.g., carbon monoxide)	Mount at approximately breathing height, considering local air movement.
Lighter than air (e.g., hydrogen)	Mount high, near the ceiling or upper structure where the gas is expected to accumulate.

Do not install GLD in locations where: the ambient temperature exceeds the specified operating range (Section 13); condensation regularly occurs; water is directly sprayed; the atmosphere is corrosive beyond the specified rating; or interfering hydrocarbon gases other than the intended target gas are continuously present. Provide sufficient clearance around the enclosure for cable access and maintenance.

6. Power Supply and Wiring

PARAMETER	VALUE	STATUS
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Input voltage	24 VDC (continuous)	Final
Source adapter (if AC-fed)	220 VAC / 50 Hz input → 24 VDC output	Final
Internal voltage rails	5 VDC (sensor/heater circuitry), 3.3 VDC (ESP32-S3 logic)	Partially available — per-rail current not yet documented separately
Measured power consumption	7.995 W @ 24 VDC (≈0.33 A)	Final (production configuration)
Cable/PSU sizing margin	≥1 A per unit	Final (sizing margin only; actual draw is well below this)
Power connector type	Not yet determined	Pending confirmation
Cable type and gauge	Not yet determined	Pending confirmation
Electrical protection	Resettable PPTC fuses (F1/F2), TVS/ESD suppression diodes, Schottky diodes on power rails	Partially available (component-level evidence)

WARNING

- Turn off the power supply before opening the enclosure or connecting/disconnecting wiring. An energized circuit may become a source of ignition in a hazardous area.
- Ground the unit in accordance with the site's electrical safety requirements before energizing.

A separate Li-ion 18650 battery pathway exists as a parallel R&D track and is not part of the standard production power configuration described above. The Ex/cell safety certification of this battery (e.g., against IEC 62133 / UN38.3) has not yet been verified and is out of scope for a standard installation using the 24 VDC supply above.

7. Communication and Network

GLD does not have a wired 4–20 mA analog output or a dry-contact relay output. Its telemetry and alarm status are transmitted wirelessly.

PARAMETER	VALUE
Radio module	E22-900MM22S (LoRa)
Operating frequency	920 MHz (regional ISM band)
Transmit power	17 dBm (firmware-configured); module maximum 22 dBm

Modulation parameters	125 kHz bandwidth / SF7 / coding rate 4/5
Link encryption	AES-128-GCM
Antenna	External, omnidirectional, SMA male, 3 dBi
Reporting interval	Configurable, default 10 seconds; alarm events are transmitted immediately
Network topology	GLD → Cluster Head (star / multi-hop mesh) → Gateway → MQTT broker → server/dashboard
Wi-Fi	Local configuration only (enabled/disabled by command). Not used for telemetry — LoRa remains the sole telemetry backbone at all times.
RS-485 / Modbus RTU interface	Read-only slave, 9600 bit/s, 8N1, Unit ID 1 (THVD1410DR transceiver). Provides 8 read-only registers: device status word, classification result, confidence, battery voltage, power source, external power indicator, LoRa transmission counter, and node ID. Intended for integration with an external controller or acquisition system; not used to control the product.

NOTE

If GLD is required to interface with a separate indicator, alarm panel, or building management system, integration should be planned through the RS-485/Modbus register set above or through the dashboard/MQTT layer — not through an analog loop or dry contact, neither of which GLD provides.

8. Precautions Before Use

CAUTION

Before energizing GLD or any connected device, recheck that all wiring connections are correct, particularly the power supply polarity and any RS-485 bus connections.

In Case of a Gas Leak Alarm

WARNING

- Follow your site's emergency response procedures immediately upon a gas leak alarm.
- If the leak occurs indoors, ventilate the area in accordance with site procedure.
- Do not operate electrical switches in or near the affected area; sparks may cause ignition.
- Confirm safety with a calibrated portable gas detector before re-entering the affected area, in accordance with site procedure.

9. Gas Detection and Alarm Operation

GLD uses eight metal-oxide gas sensor channels (MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135), with sensing elements directly exposed to ambient air. Signals are digitized by a 24-bit multi-channel ADC (ADS1256IDBR, 30,000 SPS) and sampled at a 500 ms scan interval. An on-device AI classifier running on the ESP32-S3 processor evaluates the sensor pattern and outputs one of three classes: **Clean Air, LPG, or H₂**. A dedicated SHT40 sensor provides auxiliary temperature/humidity telemetry.

Current alarm logic: an alarm is raised when the classified condition is not Clean Air and the classification confidence is at least 30%.

On alarm, two parallel channels are activated: (1) the local visual/audible alarm module on the unit itself, and (2) a push notification transmitted over the LoRa network to the central dashboard, targeting a cycle time of approximately 10 seconds. Under normal (non-alarm) conditions, the system uses a longer polling interval (approximately 90 seconds) to conserve network bandwidth.

VALIDATION STATUS

The push-alarm pathway has been successfully validated on a campus mesh test network. Validation of this pathway at an actual refinery (RU) installation has not yet been performed.

NOTE

The current 3-class model does not yet cover CO₂, Benzene, CO, or H₂S. Expansion of gas coverage depends on the availability of reference gas samples for on-site training (see Section 11, Maintenance).

10. Status Indication and Fault Conditions

GLD has no on-unit digital display or numeric fault-code readout comparable to a conventional stand-alone gas detector. Status and fault information are available through two channels:

- **Local alarm module** — visual (flashing) and audible indication of an active gas alarm.
- **Remote dashboard / RS-485 status register** — device status word, classification result, confidence, battery voltage, power source, and external power indicator, available via the central dashboard or by polling the RS-485/Modbus register set described in Section 7.

A dedicated on-unit fault/error code table, of the kind found in conventional stand-alone gas detectors, has not been defined for GLD. Fault diagnosis currently relies on the dashboard and/or the RS-485 status word; see Section 12, Troubleshooting, for the fault conditions that are understood at this time.

11. Maintenance and Inspection

11.1 Periodic Inspection

ITEM	WHAT TO CHECK
Sensing element / mesh	Clogging, corrosion, or physical damage to the sensor mesh cover.
Enclosure and mounting hardware	Corrosion of the enclosure, mounting bracket, U-bolt, and fasteners; security of the mounting.
Antenna and cabling	Integrity of the antenna, its connector, and the power/RS-485 cable entry.
Local alarm module	Visual/audible operation, physical condition of the lens and housing.
Site conditions around the unit	Confirm nothing obstructs the diffusion of gas to the sensing elements.

A formal inspection frequency (e.g., monthly/quarterly/annual, as used by comparable certified detectors) has not yet been established for GLD and should be defined by LGU together with Pertamina RU IV as part of the operational handover.

11.2 Sensor Response Verification

GLD's sensor response is verified using the project's portable gas test chamber and reference gas samples, following the project's on-site chamber-based methodology: the chamber is brought to the deployment location, a controlled gas sample is introduced, and the sensor/classifier response is checked and, where needed, the on-device model is retrained using the locally captured data. This differs from the zero/span calibration procedure used by conventional analog gas detectors, since GLD does not have an analog output to calibrate; instead, its AI classifier is verified and retrained against reference gas exposure.

11.3 Sensor Replacement and Service Life

The metal-oxide (MQ-series) sensing elements used in GLD are consumable components with a finite service life, as is typical for this sensor technology. A specific recommended replacement interval for GLD's sensors has not yet been established and should not be assumed to match the service life figures published for other sensor technologies (e.g., electrochemical or NDIR sensors used in other detector models). See Section 16, Service Life.

12. Troubleshooting

SYMPTOM	POSSIBLE CAUSE	SUGGESTED ACTION
Unit does not report telemetry to the dashboard	Power not connected/incorrect wiring; LoRa link to Cluster Head not established; Cluster Head or Gateway offline	Check the 24 VDC supply and wiring; confirm the Cluster Head and Gateway are powered and within range; check dashboard connectivity status.

Local alarm module does not activate during a known alarm condition	Alarm module wiring/connector fault; module damage	Check the alarm module cable/connector; inspect the module for physical damage.
RS-485/Modbus master does not receive a response	Bus wiring fault; incorrect baud rate/parity; incorrect Unit ID	Confirm 9600 bps, 8N1, Unit ID 1, and correct A/B/GND wiring.
Classification appears incorrect or inconsistent	Sensor mesh contaminated or degraded; interfering hydrocarbon gas present; model not trained for the gas present	Inspect and clean the sensor mesh; verify the ambient environment; confirm the gas present is within the model's current 3-class coverage (Section 9).

If a fault condition is not covered above, or does not resolve after the suggested action, contact LGU technical support.

13. Specifications

PARAMETER	VALUE	STATUS
Sampling method	Diffusion type; sensing elements directly exposed to ambient air	Final
Gas sensors	MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135 (8 channels)	Final
Auxiliary sensor	SHT40 temperature/humidity (I2C)	Final
Signal acquisition	ADS1256IDBR, 24-bit, 30,000 SPS; 500 ms sensor scan interval	Final
Processing unit	ESP32-S3-WROOM-1U-N16R8	Final
Detection / classification	On-device AI classifier, 3 classes: Clean Air, LPG, H ₂ . Does not yet cover CO ₂ , Benzene, CO, H ₂ S.	Final (current coverage)
Alarm logic	Alarm when classified condition ≠ Clean Air and confidence ≥ 30%	Final
Alarm indication	Local visual/audible module; LoRa push notification to dashboard (≈10 s target cycle)	Final (design); field validation at RU pending
Radio	LoRa, E22-900MM22S, 920 MHz, 17 dBm (fw), 125 kHz/SF7/CR4-5, AES-128-GCM	Final
Antenna	External, omnidirectional, SMA male, 3 dBi	Final

Auxiliary digital interface	RS-485/Modbus RTU, read-only, 9600 8N1, Unit ID 1	Final
Analog output / dry contact	Not provided	Final (confirmed absent)
Power supply	24 VDC continuous (from 220 VAC/50 Hz adapter or site 24 VDC feed)	Final
Power consumption	7.995 W @ 24 VDC (≈0.33 A), production configuration	Final
Dimensions (L×W×H)	200 × 90 × 290 mm	Final
Weight	≈2 kg	Final (project-confirmed, not yet calibration-verified)
Enclosure material	Aluminum alloy + stainless steel	Partially available — specific grade pending
Mounting	L-bracket + U-bolt 2"/DN50, non-invasive (no drilling/welding)	Final
Mounting orientation (vertical pole)	Not yet verified in writing	Pending confirmation
Fastener size/grade	Not yet determined	Pending confirmation
IP (ingress protection) rating	Not yet tested/determined	Pending confirmation
Cable entry / gland specification	Not yet determined	Pending confirmation
Operating temperature range	−40°C to +85°C (ambient)	Final (project-confirmed; not yet calibration/test-verified — see note below)
Operating humidity range	Not yet determined	Pending confirmation

NOTE ON OPERATING TEMPERATURE RANGE

The −40°C to +85°C figure above is a project-confirmed value corresponding to a typical extended-industrial electronics component rating. For reference, a comparable certified diffusion-type gas detector (New Cosmos KD-12/KD-12R) specifies a narrower field ambient operating range of −10°C to 50°C (RH 10–90%), which is more representative of actual refinery ambient conditions. Whether −40°C/+85°C is intended as GLD's true field deployment target, or is a component rating rather than a field specification, should be confirmed before this value is relied upon in a certification submission.

14. Certification Status

IMPORTANT

GLD is **not yet ATEX/IECEx certified**. The classification parameters below are the applicant's proposal for the purpose of the ongoing certification process; they are recorded here as a request, not as a granted or agreed classification. Do not represent GLD as certified, or as suitable for a specific hazardous-area zone, until a certificate has been issued.

PARAMETER	REQUESTED / TARGET VALUE	STATUS
Equipment category / group	Category 2G, Group II	Requested — subject to ExCB assessment
Zone	Zone 1	Internal recommendation — pending ExCB confirmation
Gas group	IIC	Internal recommendation , not an ExCB decision
Temperature class	T4 ($\leq 135^{\circ}\text{C}$)	Target — verification outstanding (no hot-spot measurement performed yet)
Type of protection	Ex d, Ex e, or Ex i — under evaluation	Open
Standards to be applied	IEC 60079-0, plus the standard corresponding to the type of protection once selected	Dependent on the item above
Certification route	IECEx Certificate of Conformity and/or ATEX EU-type examination	To be confirmed with the ExCB
Nameplate / marking	Cannot be finalized until the above parameters are confirmed by the ExCB	Not yet available

The full technical submission supporting this certification effort is maintained separately in the project's IECEx/ATEX Technical Datasheet. This Instruction Manual is intended to satisfy the "usage and installation instructions" item of that submission once reviewed and approved.

15. Warranty

A formal warranty period and terms for GLD have not yet been established. Warranty terms will be defined by LGU as part of the commercial agreement with Pertamina and are not addressed by this technical manual.

16. Service Life

An overall design service life and a specific gas-sensor replacement interval have not yet been established for GLD. As with any metal-oxide gas sensor, sensing performance is expected to degrade over time and with exposure to high-concentration or contaminating gases; the sensing elements should be treated as consumable components requiring periodic verification (Section 11) and eventual replacement, on an interval to be defined once field experience is available.

17. Glossary

TERM	MEANING
GLD	Gas Leak Detector — the sensor node described in this manual.
Cluster Head (CH)	An intermediate LoRa node that relays telemetry from one or more GLD units toward the Gateway, and can provide mesh routing between Cluster Heads.
Gateway	The device that bridges the LoRa mesh network to the MQTT broker and, from there, to the central server/dashboard.
Diffusion type	A gas detection method that relies on the natural diffusion of gas to the sensing element, rather than forced sampling.
Hazardous area	An area where an explosive gas atmosphere may be present, requiring appropriately certified equipment.
Zone	A classification of a hazardous area based on the likelihood and duration of an explosive atmosphere being present (e.g., Zone 0, 1, 2).
Gas group	A classification (e.g., IIA, IIB, IIC) of equipment suitability based on the ignition characteristics of the gases present.
Temperature class	A classification (T1–T6) of the maximum surface temperature that equipment may reach, to avoid igniting a surrounding explosive atmosphere.
LEL	Lower Explosive Limit — the lowest concentration of a flammable gas that will ignite when mixed with air.
RSSI	Received Signal Strength Indicator — used by Cluster Heads to select the best available parent link.